

Task 1: Client Adversity Profiles

Adversity of a client at horizon τ is the share of that client's trades that are adverse from the LP's perspective, i.e. $\text{side} \cdot (M_\tau - \text{TP}) < 0$ (volume is always positive so it does not affect the label). The table reports the adversity percentage for each client at every horizon.

client	tau=5	tau=10	tau=15	tau=20	tau=25	tau=30
A	39.7204	41.3958	42.1859	41.9509	41.8959	41.7133
B	40.7505	42.3591	43.1579	43.3049	43.4241	43.2965
C	41.3748	43.3486	44.2474	44.8386	45.3851	45.9225
D	43.3193	46.5217	48.1667	49.7094	50.6196	51.7529
E	44.9561	48.4875	50.9196	52.1314	53.8495	54.8449
F	47.0073	51.6674	54.2606	56.9399	59.49	61.8822

Observed Trends:

- As you can see, adversity rises monotonically with the horizon for each client. Immediately after a trade the mid sits roughly where we dealt, so the adverse fraction starts near 40–47%. As τ grows the mid drifts in the direction the client traded, so more positions would close at a loss for us. This signifies adverse selection. Informed flow predicts short-horizon mid moves.
- Clients separate cleanly into a toxicity ranking $A < B < C < D < E < F$. As you might have noticed, A, B and C are very close to each other and also all three of them barely steepen with increase in τ , this is uninformed liquidity that we can internalise profitably. D is a transition case. E and F climb sharply (F reaches ~62% at $\tau = 30$), so they are costly to hold.
- The slope of the profile is itself the toxicity signal. A flat profile means the client's trades are uninformative about future mid moves, whereas a steep upward profile means the longer we keep the position, the more it hurts. These are precisely the clients for whom externalisation (Tasks 3–4) and wider quotes (Task 2) are most valuable.